Thermal Behaviour of Layered Composites

A short report used as a layout fixture

Abstract

This report summarises a measurement campaign on layered composite plates exposed to cyclic heating. The plates were held at 180 C for 20 minutes and cooled in still air. Deflection was recorded at the centre of each plate.

1. Method

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane. Torque on the clamping bolts was set to 12 Nm +/- 0.5 Nm, which the supplier gives as the limit for this fixture.

Temperature was logged every 2 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here.

Figure 1: mean deflection against cycle count.

2. Results

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles.

The table below lists the mean deflection per plate. Values are millimetres and must be read as measured; they are not rounded.

Plate	Cycles	Deflection
A-1	10	0.42 mm
A-2	20	0.45 mm
B-1	10	0.61 mm
B-2	20	0.63 mm

3. Conclusion

The fixture holds the specimens without measurable creep at 12 Nm. A longer campaign is needed before the result is extended to thicker stacks.